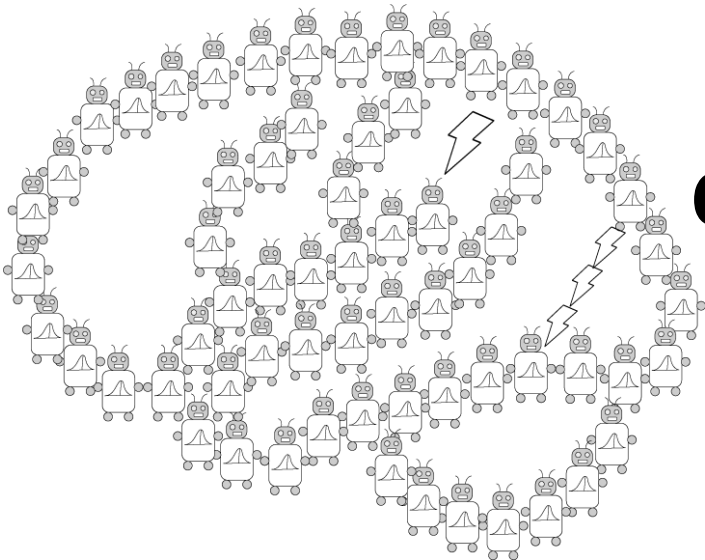


Markov Chains and Networks



**PSYCH/COMP SCI/BMI 841:
Computational Cognitive Science**

Joseph Austerweil

March 6, 2025

- Admin Matters
- Markov Chains
- Network Theory
- **Short rest 1**
- Student presentations:
 - Karl presenting Muhkerjee et al (2017)
 - Bujji presenting Griffiths et al. (2007)
- **Short rest 2**
- Student presentations:
 - Aakash + Ashwini presenting Abbott et al. (2012)
 - Caleb presenting Zemla and Austerweil (2019)

Administrative Matters

- Caught up on last week's discussion posts, but now that I'm delayed, it will take a bit to catch up.
- Generalization Assignment was due yesterday at 8:59pm
 - Remember you have three “free” late days for assignments.
 - Late days do not prorate (used up in 24-hour chunks).
 - Some folks are still turning this in, so we'll chat about it next week.
- Project proposal due next Wednesday Mar 12 8:59pm
 - No late days for project proposal (or final paper/presentation)
- Assignment 3 will be going up in the next week.
- Questions?

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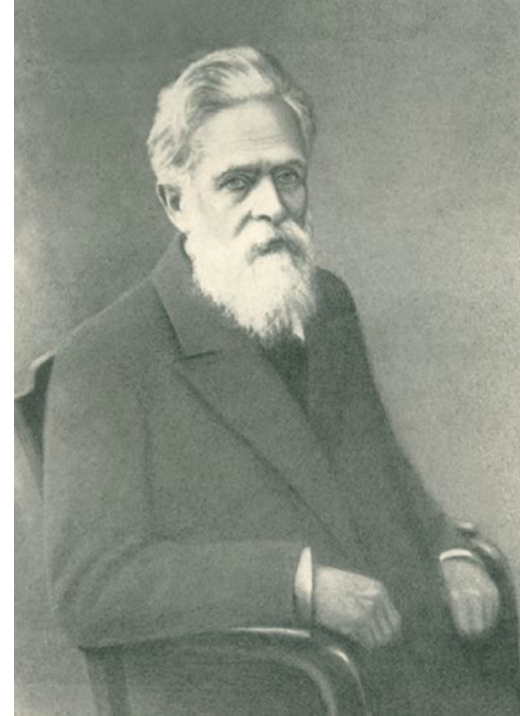
Markov



Really Past



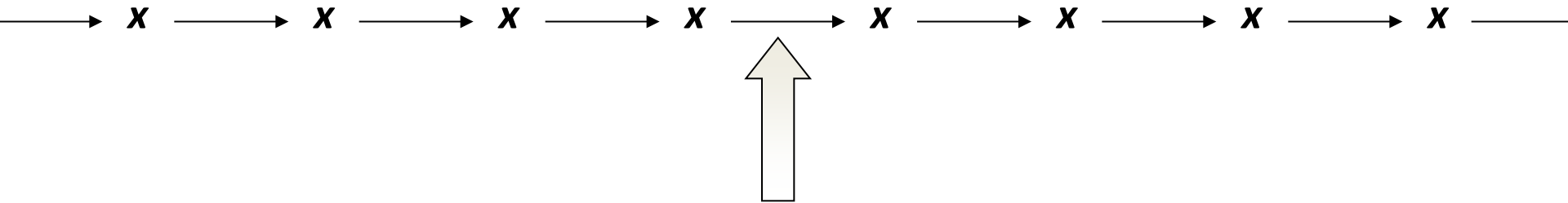
Past



Less Past

Given the present, the future is independent of the past.

Markov chains



Transition matrix

$$\mathbf{P} = P(\mathbf{x}^{(t+1)} | \mathbf{x}^{(t)})$$

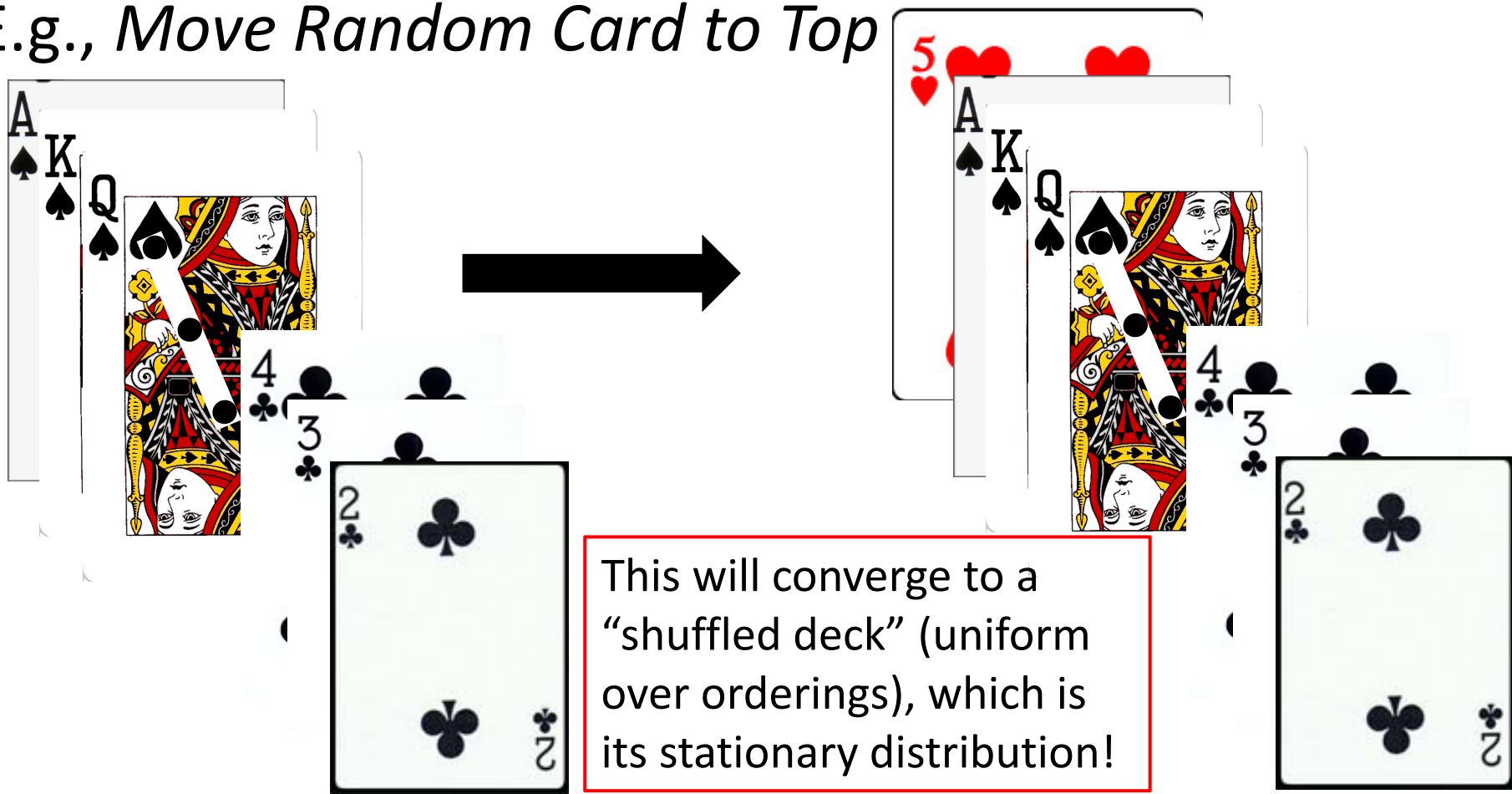
Variables $\mathbf{x}^{(t+1)}$ independent of history given $\mathbf{x}^{(t)}$

Converges to a *stationary distribution* in most situations (i.e., if it is *ergodic*).

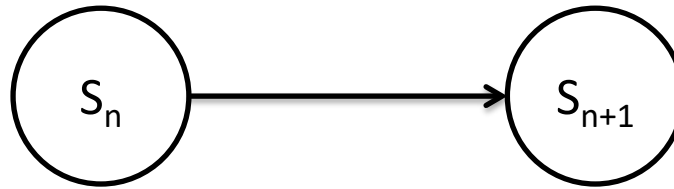
Example: Card Shuffling

A *shuffle* is a transition matrix of a Markov chain, whose states are the order of cards in a deck.

E.g., *Move Random Card to Top*

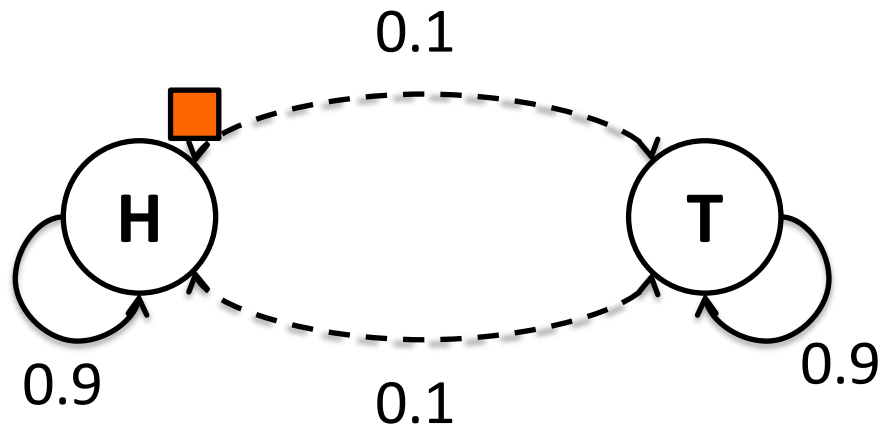


Another look at Markov chains/processes



Transition
Matrix
view

Finite state automaton (machine) view



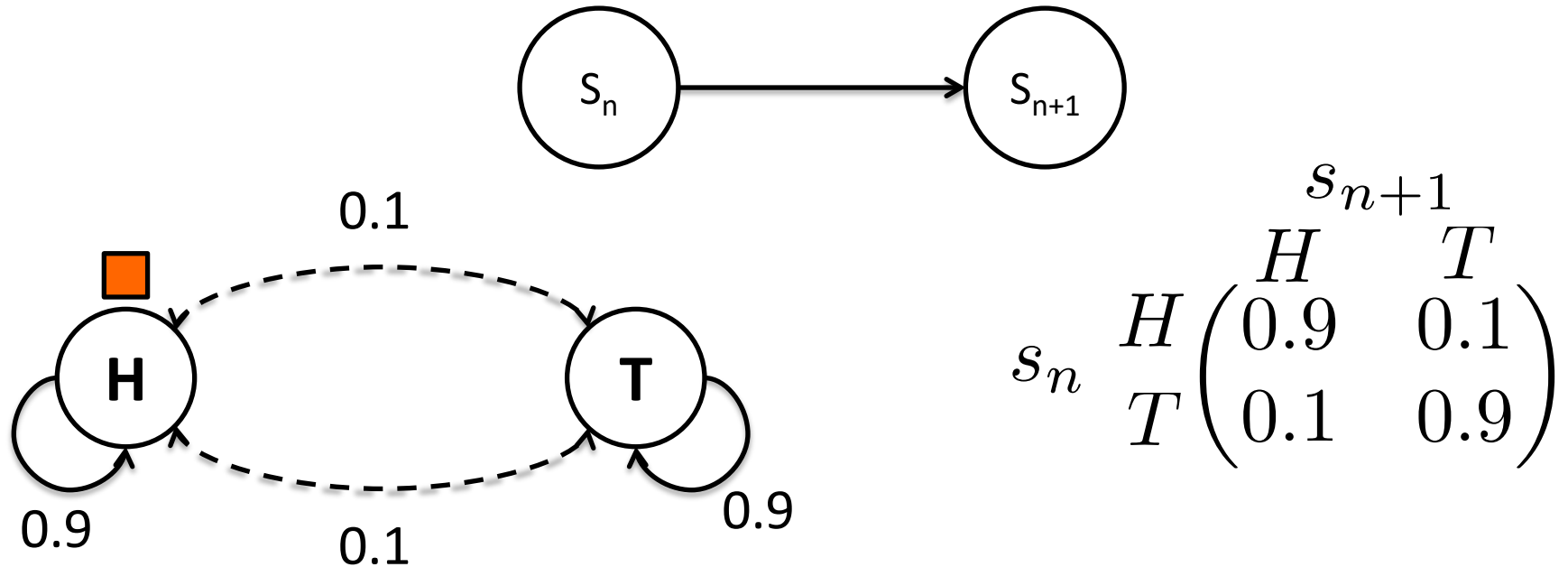
		s_{n+1}	
		H	T
s_n	H	0.9	0.1
	T	0.1	0.9

$s_0 = H$ Sample $U[0,1] = 0.56 < 0.9 \Rightarrow s_1 = H$

$s_1 = H$ Sample $U[0,1] = 0.37 < 0.9 \Rightarrow s_2 = H$

$s_2 = H$ Sample $U[0,1] = 0.96 > 0.9 \Rightarrow s_3 = T$

Another look at Markov chains/processes

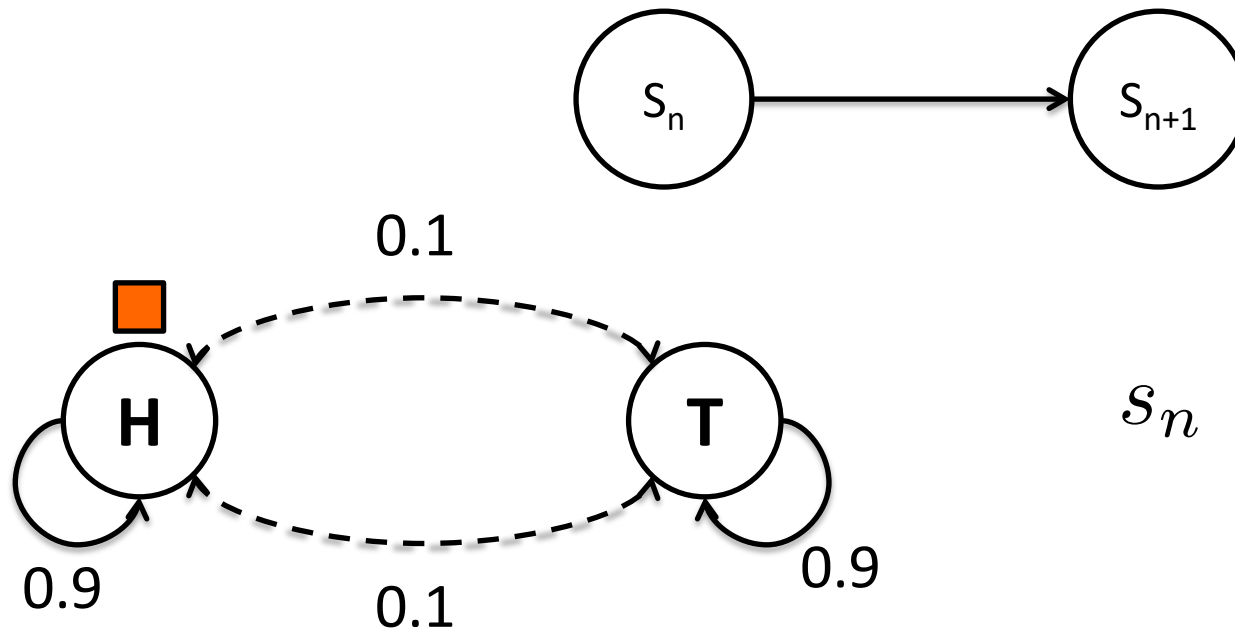


Samples from this process (starting at H):

HHHHHHHTTTTTTTTTTTT
 HHHHHHHHTTTTTTHHHTHHH
 HHHHHHHHTTTTTTTTTTTT
 HHHHTTTTHHHHHHHHHHHH
 HHHHHHHHTTTTTTTTTTTT

HHHHHHHHHTTTTHHHHHHHH
 HHHHHHHHTTTTTTTTTTTT
 HHHHHHHHTTTTTTTTTTTT
 HHHHHHTTTTTTTTTTTTTT
 HHTTTTTTTTTTTTTTTTTT

Another look at Markov chains/processes



$$s_n \begin{matrix} & s_{n+1} \\ \begin{matrix} H \\ T \end{matrix} & \begin{pmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{pmatrix} \end{matrix}$$

What's the stationary distribution? (intuitively $\frac{1}{2}$, but how to see?)

$$\mathbf{P}^2 = \begin{pmatrix} 0.82 & 0.18 \\ 0.18 & 0.82 \end{pmatrix} \quad \mathbf{P}^6 = \begin{pmatrix} 0.63 & 0.37 \\ 0.37 & 0.63 \end{pmatrix} \quad \mathbf{P}^{15} = \begin{pmatrix} 0.52 & 0.48 \\ 0.48 & 0.52 \end{pmatrix}$$

$$\mathbf{P}^4 = \begin{pmatrix} 0.70 & 0.30 \\ 0.30 & 0.70 \end{pmatrix} \quad \mathbf{P}^8 = \begin{pmatrix} 0.58 & 0.42 \\ 0.42 & 0.58 \end{pmatrix} \quad \mathbf{P}^{20} = \begin{pmatrix} 0.51 & 0.49 \\ 0.49 & 0.51 \end{pmatrix}$$

Multi-step Probability and the Stationary Distribution

P_{ij}^T is the probability of starting in state i and ending in j after T steps

What happens if I multiply $\mathbf{P}$ by itself a lot?

$$\vec{\pi}\mathbf{P} = \lambda\vec{\pi}$$

Largest Eigenvector

π_i is the relative frequency of visits to state i by the Markov chain after many steps.

Largest eigenvalue is 1.

Property of “stochastic matrices”

Note on Eigenvectors

Eigenvectors of a matrix $\mathbf{X}$ are those vectors that are “unchanged” when multiplied by $\mathbf{X}$.

They only differ from the original by a **scalar**.

That scalar is the eigenvector’s eigenvalue.

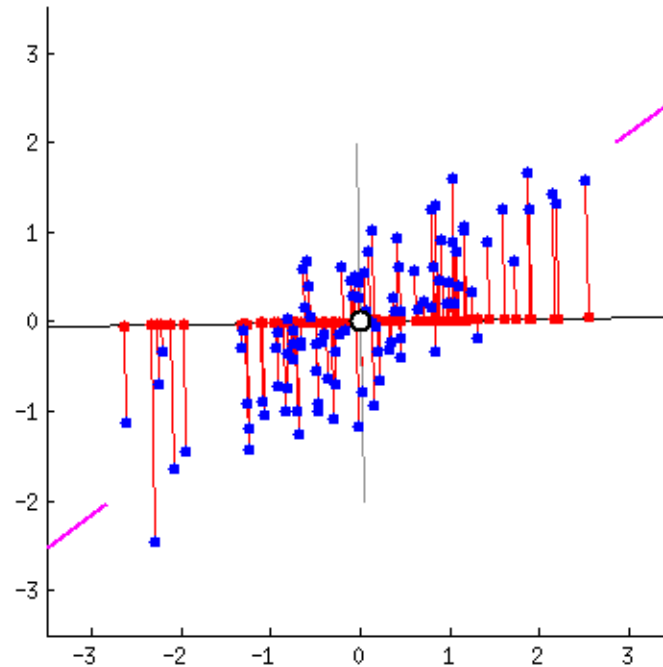
Eigenvectors and eigenvalues are very useful descriptions of a matrix.

The sum of the eigenvalues is equal to the sum of the diagonal of a matrix (called trace).

The product of eigenvalues is equal to the determinant.

Note: Square matrices with positive values (more generally, **positive semidefinite**) have **real eigenvalues**.

Visual interpretation



Blue dots = data

Black line = line centered at data with varying slope

Red lines/points = Best fitting points for data where data must live on black line.

Magenta = largest eigenvector – blue points “spread” on it the most AND has the smallest reconstruction error (minimal sum of red lines).

From <https://stats.stackexchange.com/questions/2691/making-sense-of-principal-component-analysis-eigenvectors-eigenvalues>

Note on Eigenvectors

Eigenvalue decomposition describes a matrix with its eigenvectors and eigenvalue.

Let $\mathbf{A}$ be a square matrix.

Let $\mathbf{\Lambda}$ be a diagonal matrix of $\mathbf{A}$'s eigenvalues.

Let $\mathbf{Q}$ be a matrix where each column is an eigenvector of $\mathbf{A}$.

$$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{-1}$$

Eigenvector decomposition of the covariance matrix of data is **principal component analysis** (including only the top k eigenvectors – those with the k largest eigenvalues).

Note on Eigenvectors

Eigenvector decomposition of the covariance matrix of data is **principal component analysis** (including only the top k eigenvectors – those with the k largest eigenvalues).

Singular value decomposition of data is equivalent to PCA on its covariance matrix.

If X is our data, singular value decomposition (SVD) combines the eigenvector decomposition of XX^T and X^TX .

Remember that SVD is the non-probabilistic version of topic modeling (latent semantic analysis).

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(Semantic) Networks

Represent words and concepts as nodes/vertices in a **graph** G , with edges indicating relationships.

$$G = (V, E) \qquad e \in E = (v_1, v_2)$$

Can be

$$\text{undirected} \quad (v_1, v_2) = (v_2, v_1)$$

$$\text{directed} \quad (v_1, v_2) \Rightarrow v_1 \rightarrow v_2 \neq (v_2, v_1)$$

Edges can be weighted $w(e_i) \in \mathbb{R}^+$

Different kinds of networks:

- inheritance hierarchies (Collins & Quillian, 1969)
- associative networks (Collins & Loftus, 1975)

From graphs to probabilities

Define a random walk on a graph from its edges:

1. Define matrix $\mathbf{L}$ that encodes edge information.

If there's an edge from node i to node j , let $L_{ij} = 1$.

You can have other weights instead of just 1.

(they do have to be positive though).

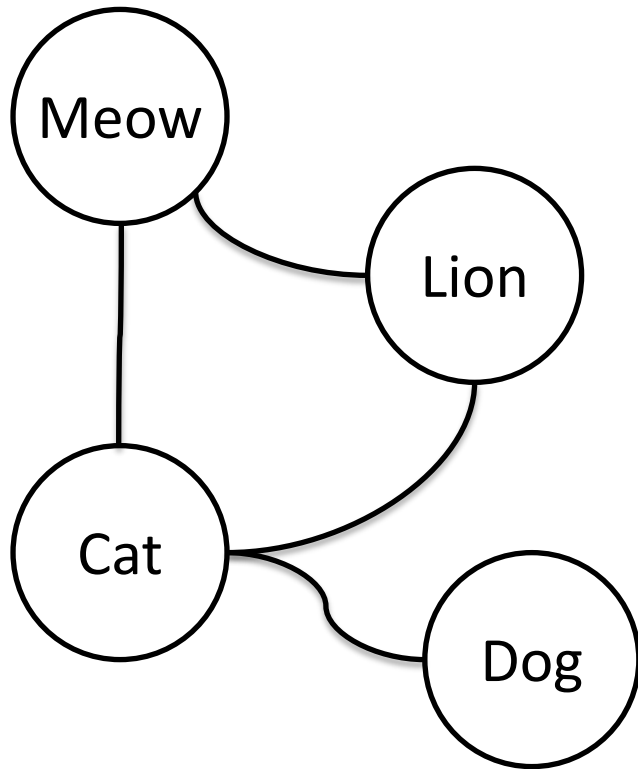
If a graph is symmetric then $L_{ij} = L_{ji}$

2. Define matrix $\mathbf{P}$ that converts $\mathbf{L}$ to the transition matrix of a RW on the graph.

$$P_{ij} = \frac{L_{ij}}{\sum_k L_{ik}}$$

The sequence of states visited by a random walk form a Markov chain

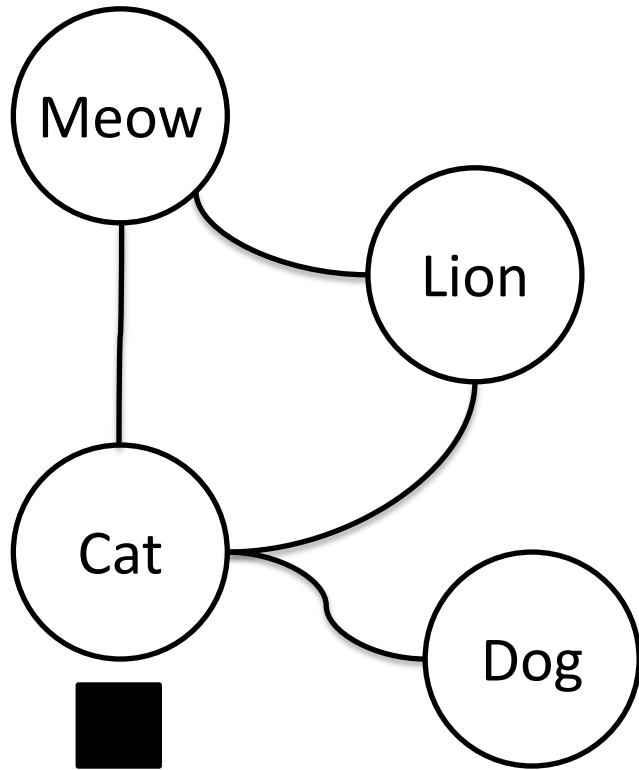
Example: Creating a random walk



$$\mathbf{L} = \begin{matrix} & \begin{matrix} \text{Meow} & \text{Lion} & \text{Cat} & \text{Dog} \end{matrix} \\ \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} & \begin{matrix} \text{Meow} \\ \text{Lion} \\ \text{Cat} \\ \text{Dog} \end{matrix} \end{matrix}$$

$$\mathbf{P} = \begin{matrix} & \begin{matrix} \text{Meow} & \text{Lion} & \text{Cat} & \text{Dog} \end{matrix} \\ \begin{pmatrix} 0 & 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 1/3 & 1/3 & 0 & 1/3 \\ 0 & 0 & 1 & 0 \end{pmatrix} & \begin{matrix} \text{Meow} \\ \text{Lion} \\ \text{Cat} \\ \text{Dog} \end{matrix} \end{matrix}$$

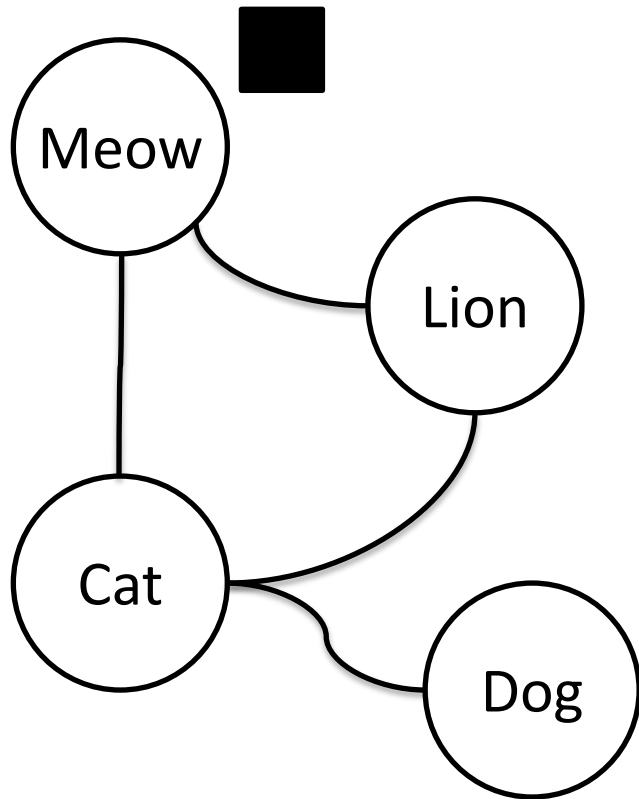
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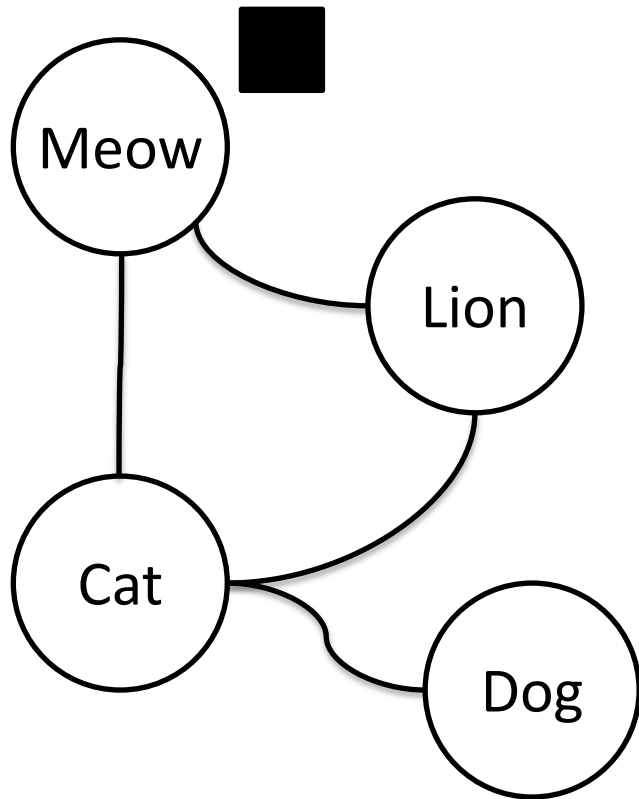


Walk visits Cat

$$\mathbf{L} = \begin{matrix} & \begin{matrix} \text{Meow} & \text{Lion} & \text{Cat} & \text{Dog} \end{matrix} \\ \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} & \begin{matrix} \text{Meow} \\ \text{Lion} \\ \text{Cat} \\ \text{Dog} \end{matrix} \end{matrix}$$

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Example: Creating a random walk

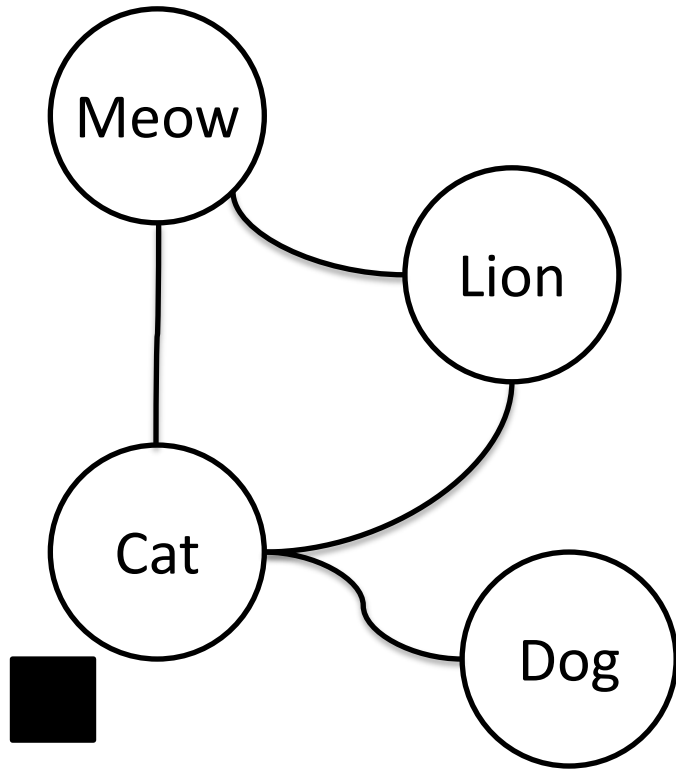


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Walk visits Cat, Meow

$$\mathbf{P} = \begin{matrix} & \begin{matrix} \text{Meow} & \text{Lion} & \text{Cat} & \text{Dog} \end{matrix} \\ \begin{pmatrix} 0 & 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 1/3 & 1/3 & 0 & 1/3 \\ 0 & 0 & 1 & 0 \end{pmatrix} & \begin{matrix} \text{Meow} \\ \text{Lion} \\ \text{Cat} \\ \text{Dog} \end{matrix} \end{matrix}$$

Example: Creating a random walk

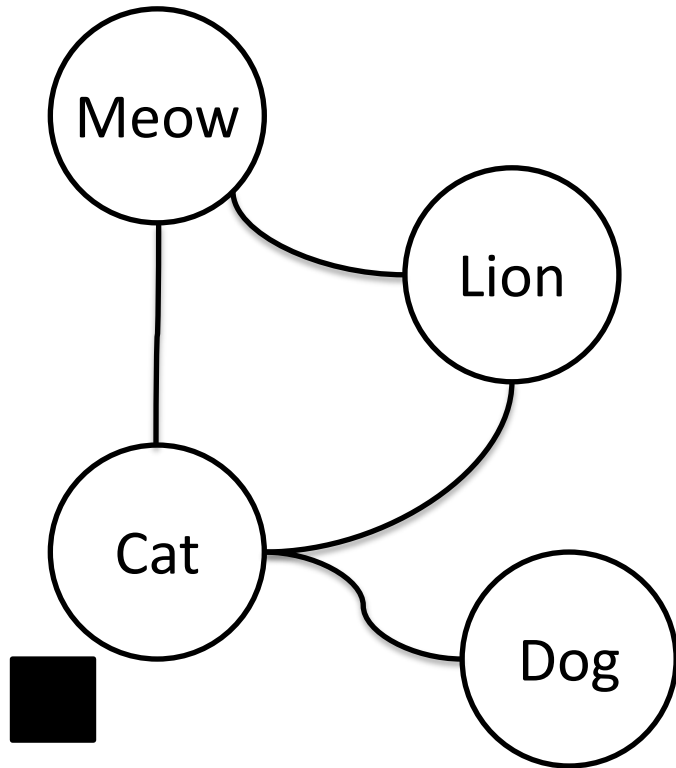


Walk visits Cat, Meow

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Example: Creating a random walk

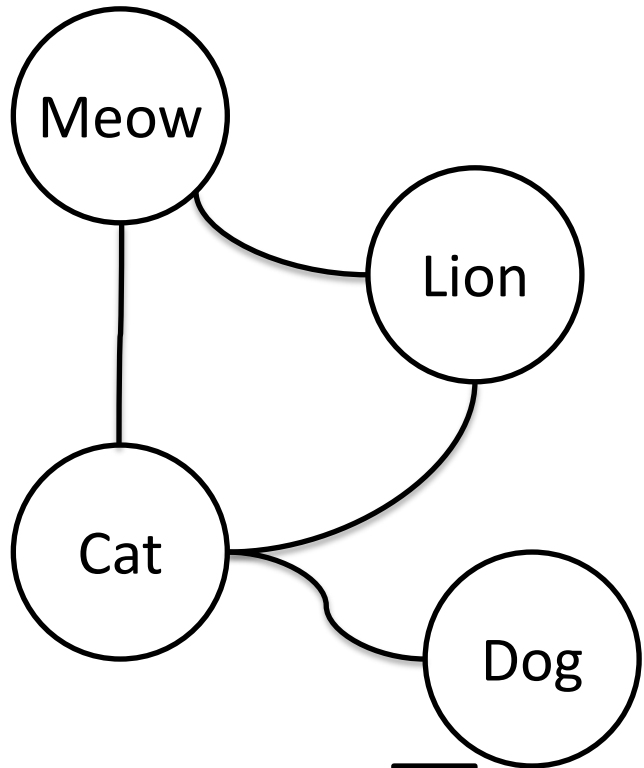


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Walk visits Cat, Meow, Cat

Example: Creating a random walk



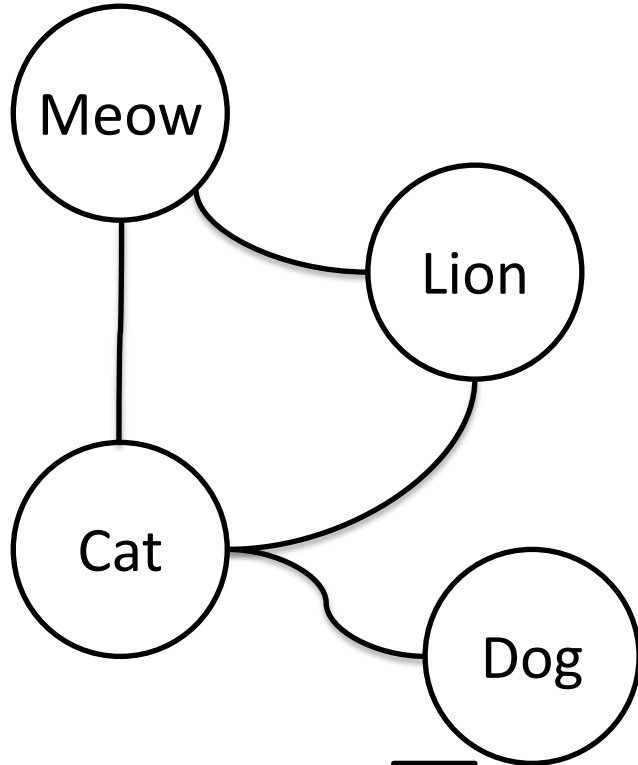
$\mathbf{P} =$

Walk visits Cat, Meow, Cat

$$\mathbf{L} = \begin{matrix} & \begin{matrix} \text{Meow} & \text{Lion} & \text{Cat} & \text{Dog} \end{matrix} \\ \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} & \begin{matrix} \text{Meow} \\ \text{Lion} \\ \text{Cat} \\ \text{Dog} \end{matrix} \end{matrix}$$

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Walk visits Cat, Meow, Cat, Dog, ...

Stationary distributions

How often will the RW visit each node?

This is the stationary distribution of the RW.
(remember RW is a Markov model)

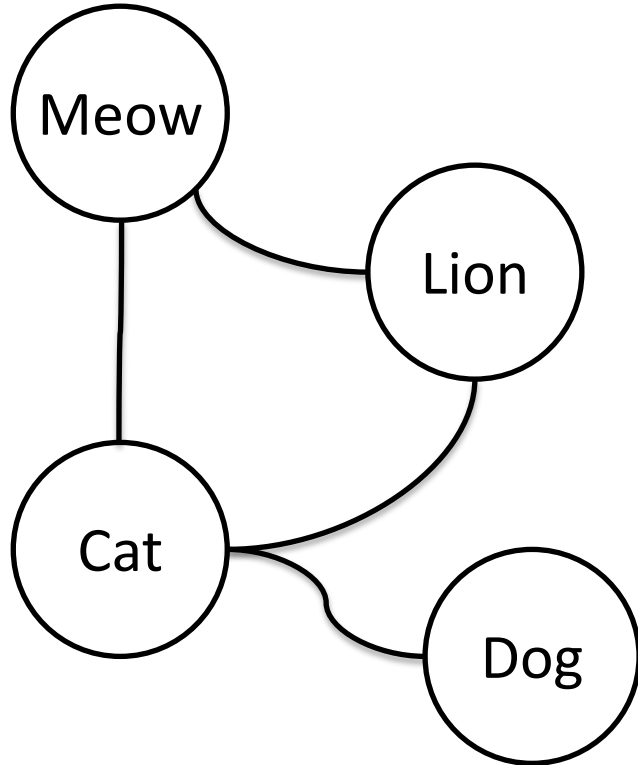
The stationary distribution π of a RW on an undirected graph is

$$\pi_i \propto \sum_k L_{ki}$$

In the case of an unweighted graph,

π_i is proportional to the # of edges connected to i
(this is also called the node's degree)

Example



$$\mathbf{P} = \begin{pmatrix} 0 & 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 1/3 & 1/3 & 0 & 1/3 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

P =

0	0.5000	0.5000	0
0.5000	0	0.5000	0
0.3333	0.3333	0	0.3333
0	0	1.0000	0

$$\vec{\pi} \propto (2 \quad 2 \quad 3 \quad 1)$$

$$\vec{\pi} = (2/8 \quad 2/8 \quad 3/8 \quad 1/8)$$

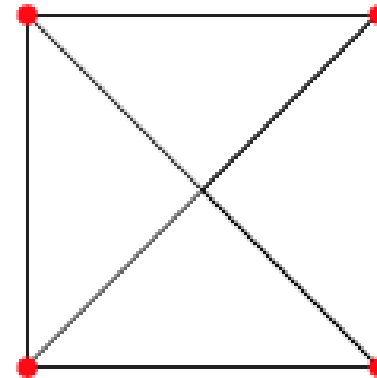
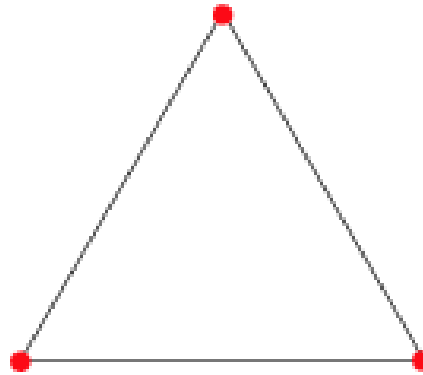
If time, more network theory

Some network terms

Degree

(d)

(d)



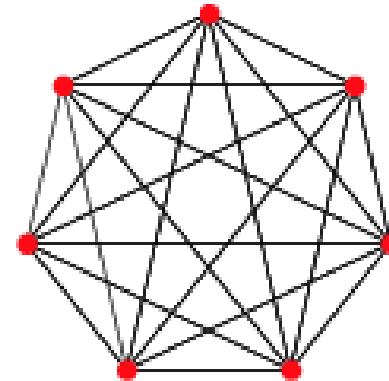
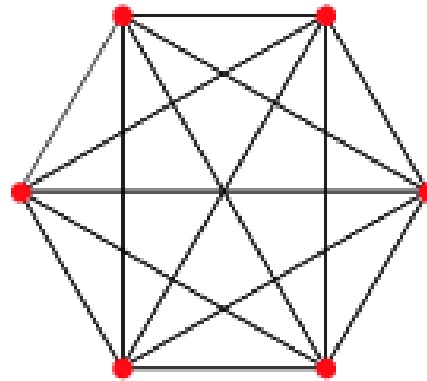
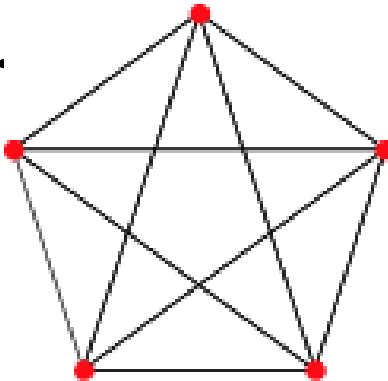
K_2

K_3

K_4

Complete

edge.



K_5

K_6

K_7

simple

Some network terms

Degree of node i : # of edges including i

(directed) in-degree: # of edges into i

(directed) out-degree: # of edges out of i

Complete graph: Graph containing every possible edge. If n nodes, n choose 2 edges.

$$\frac{1}{2}n(n - 1)$$

Edge Density: # of edges in graph compared to complete graph.

A graph is **sparse** if its # of edges $\ll$ to # in complete graph.

Some network terms

Shortest path length (Geodesic distance):

The # of edges on the shortest path between two nodes on a network.

For weighted edges, it is the path with minimal sum of edge weights.

Average shortest path length:

Average of the shortest path between every pair of nodes on a network.

Diameter: Largest path length

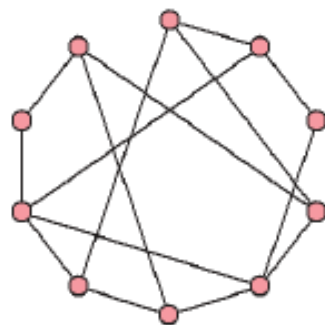
Some network terms

Clustering coefficient:

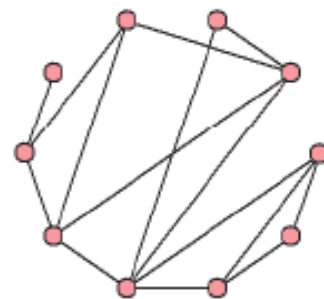
How “clustered” is the network?

(Global):
$$C = \frac{3 \times \text{number of triangles}}{\text{number of triplets of nodes}}$$

(a)

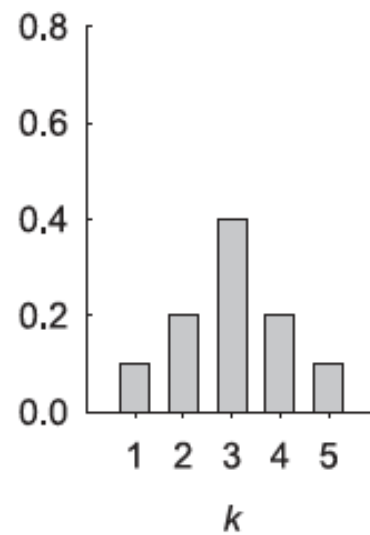
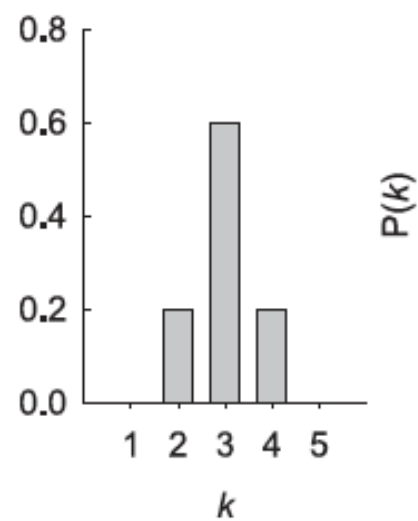


$$\begin{aligned}n &= 10 \\ \langle k \rangle &= 3 \\ L &= 1.8 \\ D &= 3 \\ C &= 0\end{aligned}$$



$$\begin{aligned}n &= 10 \\ \langle k \rangle &= 3 \\ L &= 2.18 \\ D &= 5 \\ C &= 0.56\end{aligned}$$

(b)



Some network terms

Clustering coefficient:

How “clustered” is the network?

(Global):
$$C = \frac{3 \times \text{number of triangles}}{\text{number of triplets of nodes}}$$

Small-world network:

Avg. shortest distance is proportional to the logarithm of the number of nodes.

Large clustering coefficient and small average shortest path length.

Some network terms

Small-world network:

Avg. shortest distance is proportional to the logarithm of the number of nodes.

Large clustering coefficient (C) and small average shortest path length (L).

$$\sigma = \frac{C/C_r}{L/L_r}$$

L_r and C_r are the statistics for “equivalent random network”. What does that mean?

Canonical Random Network: Erdős-Rényi

A Erdős-Rényi network is a network formed from n nodes and m edges where the m edges are assigned at random to pairs of nodes.

A variant instead defines it in terms of number of nodes and probability of an edge between any two nodes.

Who cares?

These networks cannot be sparse and clustered.

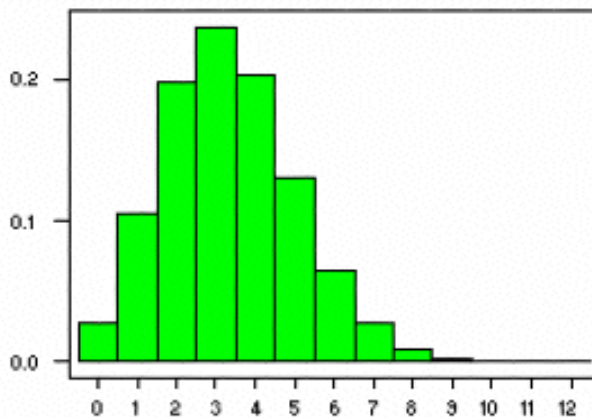
Degree distributions

The degree distribution of a network is a histogram of the degree for each node.

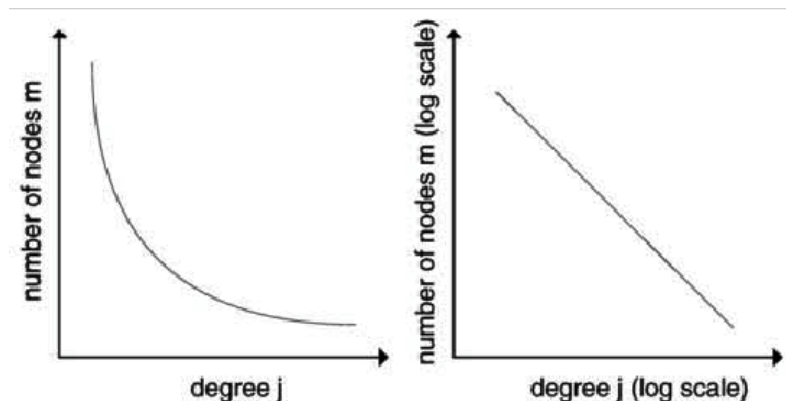
Erdős-Rényi have Binomial(n, p) degree distribution.

Real-world networks have *scale-free* or *power-law* degree distributions.

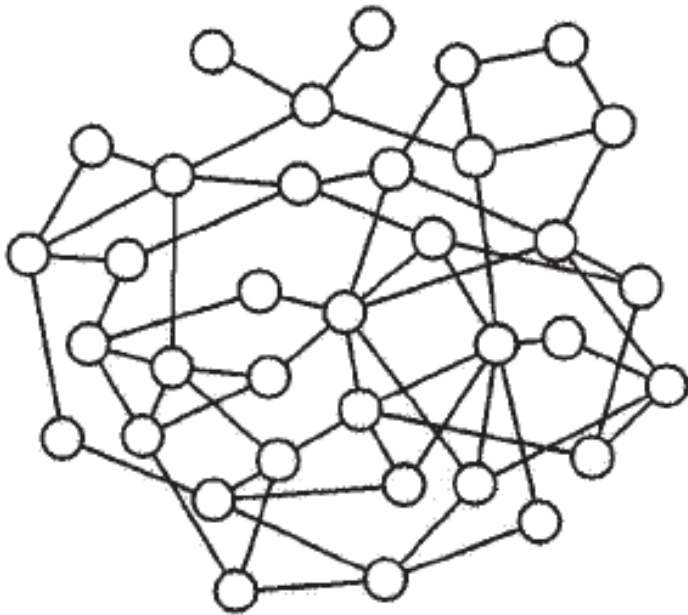
Binomial



Scale-free/Power-law



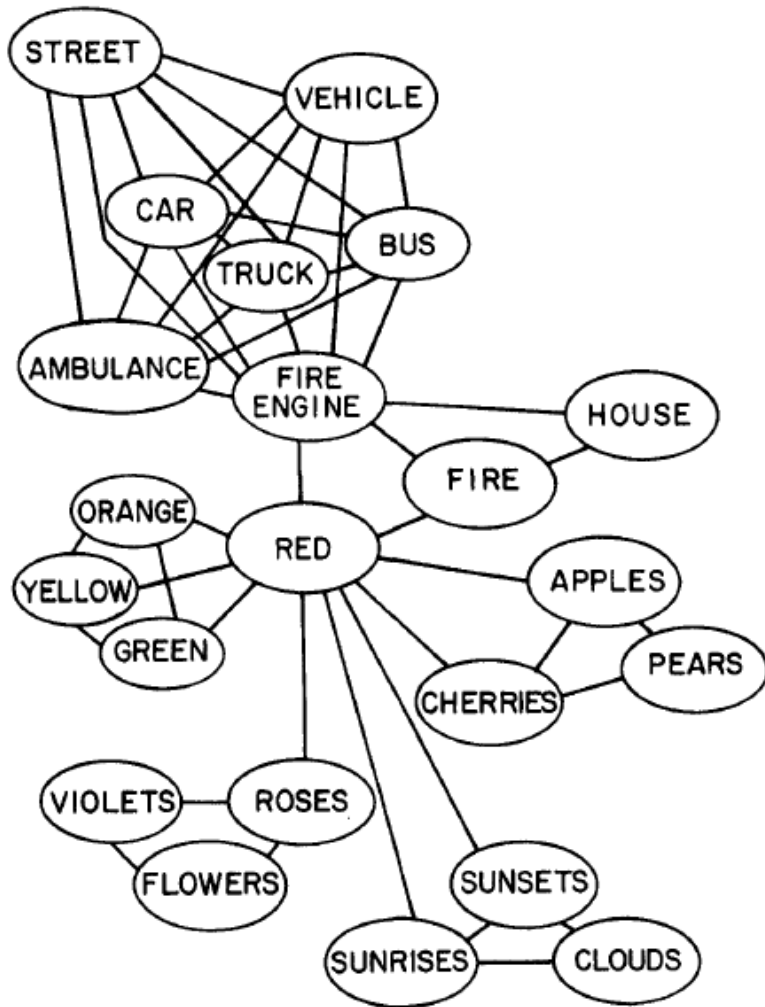
Erdos-Renyi vs. Power-Law



Network representations: Semantic/Associative Networks

Links represent associations between words or concepts.

Links may or may not be weighted.



Properties of semantic networks

USF Association Network: 5,048 nodes; ~ 63,000 edges
(of ~6 million possible)

Sparse Network contains only a small number of the possible edges between nodes.

For USF network, avg. # of edges per node is 11!

“Small world” The shortest path between any two words tends to be small.

The avg. shortest path between any two nodes is 3!

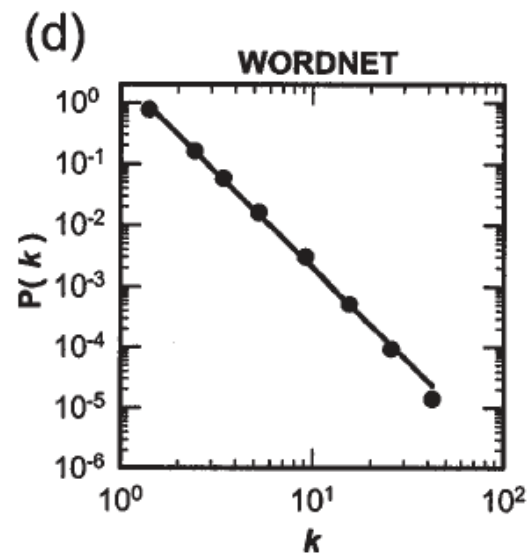
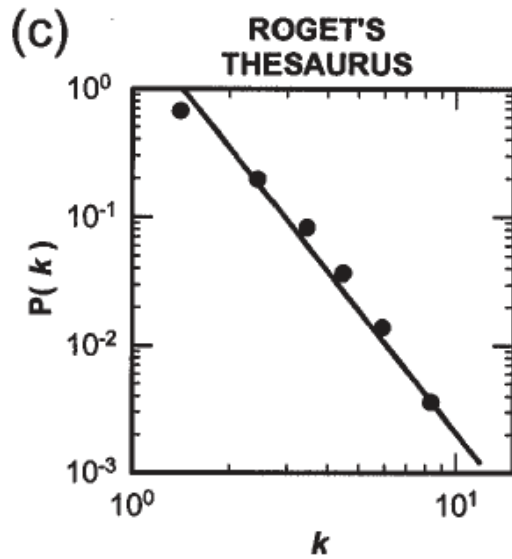
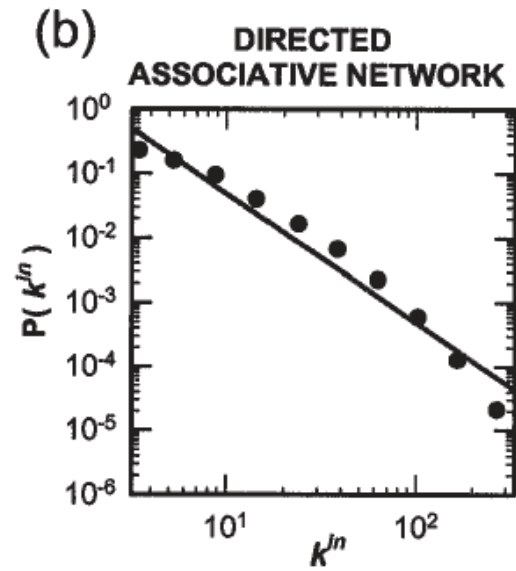
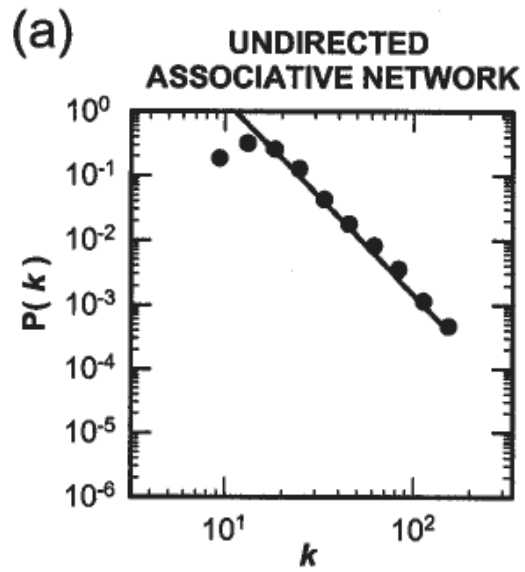
Power-law degree Most words have few edges, but a few have many edges.

Clustering Neighbors of a node are likely to be connected by an edge.

(Nelson, McEvoy & Schreiber, 1998; Steyvers & Tenenbaum, 2005)

Variable	Type	Associative Network		Roget	WordNet
		Undirected	Directed		
n	words	5,018	5,018	29,381	122,005
	classes	—	—	1,000	99,642
$\langle k \rangle$	words	22.0	12.7	1.7	1.6
	classes	—	—	49.6	4.0
L		3.04	4.27	5.60	10.56
D		5	10	10	27
C		.186	.186	.875	.0265
γ		3.01	1.79	3.19	3.11
L_{random}		3.03	4.26	5.43	10.61
C_{random}		4.35E-03	4.35E-03	.613	1.29E-04

Note. n = the number of nodes; $\langle k \rangle$ = the average number of connections; L = the average shortest path length; D = the diameter of the network; C = clustering coefficient; γ = power law exponent for the distribution of the number of edges in undirected networks and incoming connections in directed networks; L_{random} = the average shortest path length with random graph of same size and density; C_{random} = the clustering coefficient for a random graph of same size and density.

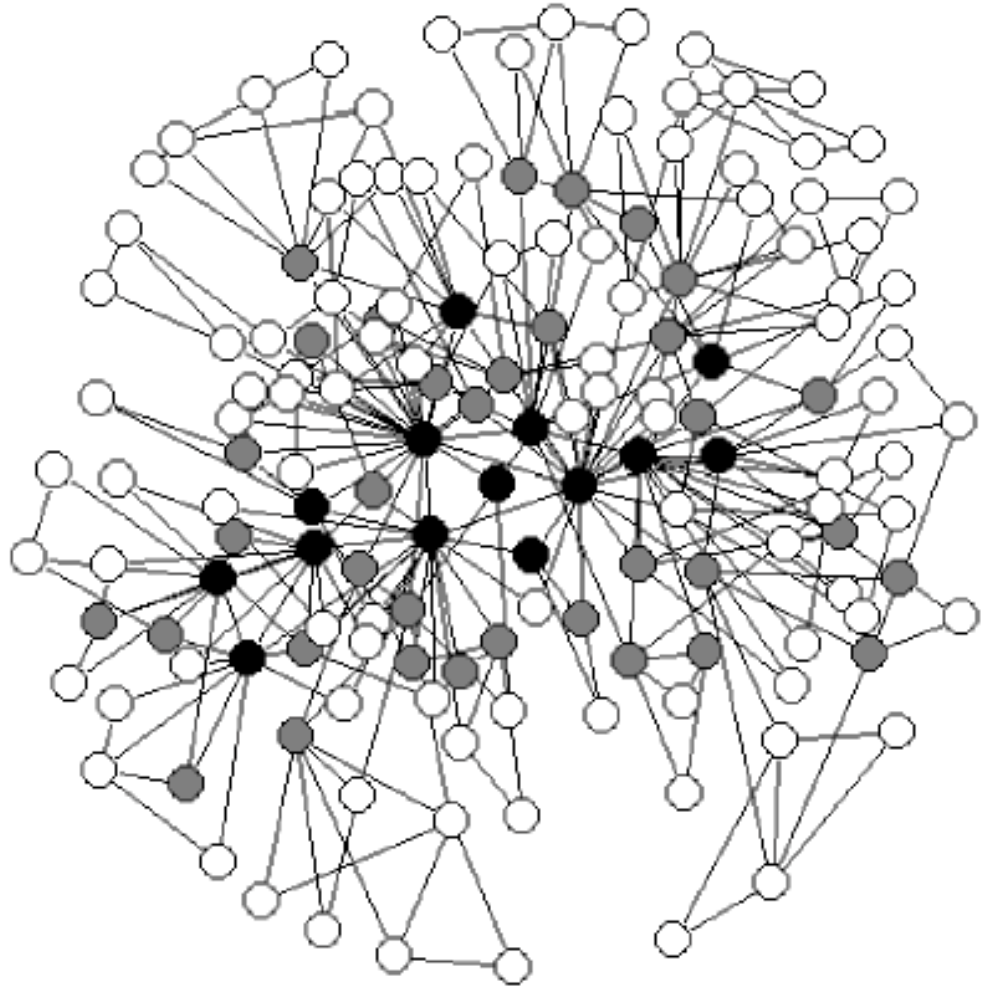


What network growth mechanisms result in scale-free, sparse, etc?

Not Erdos-Renyi.

One method (similar to Barabasi & Albert, 1999)

New nodes connect to existing nodes proportional to their degree.



- Admin Matters
- Markov Chains
- Network Theory
- **Short rest 1**
- Student presentations:
 - Karl presenting Muhkerjee et al (2017)
 - Bujji presenting Griffiths et al. (2007)
- **Short rest 2**
- Student presentations:
 - Aakash + Ashwini presenting Abbott et al. (2012)
 - Caleb presenting Zemla and Austerweil (2019)